LESSON 7.12

DIVIDING INTEGERS USING INTEGER CHIPS

LESSON INTRODUCTION

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

LEARNING TARGETS

- I can use integer chips and patterns to discover rules for dividing integers.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- How can you use integer chips to divide integers?
- How are multiplication and division of integers related?
- What patterns are there with the sign of the quotient of integers?

LESSON NARRATIVE

In this lesson, students learn how to divide integers. Students relate division to repeated subtraction and model division with integer chips. Students then relate the generalizations they can make about dividing integers to their previous learning multiplying integers. Students discover patterns with the signs of quotients of integers.

COMPONENT SUMMARY

7.12.1 WARM-UP *(~5 MIN.)*

Notice and wonder about an image of a double negative

7.12.2 GUIDED INSTRUCTION *(10 MIN.)*

Use integer chips to represent division

7.12.3 COLLABORATION *(5 MIN.)*

Collaboratively consider how integer multiplication and division are related

7.12.4 BRING IT TOGETHER *(5 MIN.)*

Consolidate how multiplication and division are related

7.12.5 COLLABORATION *(10 MIN.)*

Determine which division expressions do not belong with a partner

7.12.6 GUIDED PRACTICE *(5 MIN.)*

Practice dividing integers

7.12.7 YOUR TURN! *(5 MIN.)*

Collaboratively write several division expressions that all result in a given quotient

7.12.8 WRAP-UP *(~5 MIN.)*

Rate level of confidence with understanding and using rules for multiplying and dividing integers

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Quotient
- Divisor
- Dividend

MATERIALS

- Integer chips

ADDITIONAL PREPARATION

- Create an anchor chart with instructions, examples, and visual models.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have trouble understanding why the divisor must be a positive integer.
- Students may struggle to represent division expressions using integer chips.
- Students may struggle to relating division equations to multiplication equations.

THINGS TO CONSIDER

- Consider creating an anchor chart to illustrate how multiplication and division of signed integers are related for students to reference.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

7.12.3 Collaboration provides an opportunity for students to discuss connections they are noticing and wondering about between the rules for dividing integers and the rules for multiplying integers.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

7.12.3 Collaboration and 7.12.4 Bring It Together provide opportunities for students to engage in mathematical discussions comparing and connecting the rules for multiplication and division of integers. Students' arguments are based on the division and multiplication problems they have previously modeled and solved.

DIFFERENTIATE INSTRUCTION

Students are exploring the relationship between multiplication and division throughout this lesson. Calculation is not the focus of this lesson, but some students may need additional support when dividing integers if they are not confident in their math facts.

Consider utilizing On-Ramp to 6th Grade learning tool to provide scaffolding for students in need of additional support.

- A Fraction Is a Quantity Formed by Its Parts
- Evaluate Expressions

7.12.1 WARM-UP: NOTICE AND WONDER

Component Overview: *Students write what they notice and wonder about a sale sign displaying a double negative. If students do not recognize the issue with the image, have them model the problem on a number line or with integer chips.*

ENRICHMENT

Does the image represent an increase or a decrease in the price? How do you know?

7.12.1 Warm-Up: Notice and Wonder

1. **Write down one thing you notice about the picture.**
   Answers will vary. I notice there is a negative sign next to the $1.00.

2. **Write down one thing you wonder about the picture.**
   Answers will vary. I wonder what "–$1.00 off" means.

3. **Isaac described the image as a math fail. Why do you think he described it that way?**
   Answers will vary. I think he described it as a math fail because –$1.00 off doesn't make sense, it would actually be a double negative which would mean that the price is actually $1.00 more!

7.12.2 Guided Instruction: Using Integer Chips to Represent Division

Recall that division is repeated subtraction.

For example, $24 \div 8$ can be expressed as $24 - 8 - 8 - 8$ because 3 groups of 8 can be subtracted from 24.

This shows that 24 is made up of 3 groups of 8.

The expression $24 \div 8$ can also be interpreted as dividing 24 into 8 equal-sized groups. In the representation shown, 24 positive integer chips are divided into 8 equal-sized groups.

+ + +	+ + +	+ + +	+ + +
+ + +	+ + +	+ + +	+ + +

The quotient is the result of the division, or the value of the integer chips in each group.

$$24 \div 8 = 3$$

1. Complete the table to describe the division, then draw the visual model using integer chips, and determine the quotient.

Division Expression	Written Description	Visual Model of Division	Quotient
$\frac{15}{5}$	Divide 15 into 5 equal-sized groups	+ + + + + + + + + + + + + + +	3
$-6 \div 2$	Divide −6 into 2 equal-sized groups	− − − − − −	−3

When modeling division with integer chips, the divisor must be positive, as this indicates the number of groups needed. When the divisor is negative, the expression can be rewritten using opposites.

For example, $\frac{8}{-2}$ can be rewritten as $\frac{-8}{2}$ because $\frac{8}{-2} = \frac{-8}{2}$.

2.

Division Expression	Equivalent Division Expression	Written Description	Visual Model of Division	Quotient
$\frac{9}{-3}$	$\frac{-9}{3}$	Divide −9 into 3 equal-sized groups	− − − − − − − − −	−3
$-12 \div -2$	$\left(\frac{12}{2}\right)$	Divide 12 into 2 equal-sized groups	+ + + + + + + + + + + +	6

7.12.2 GUIDED INSTRUCTION: USING INTEGER CHIPS TO REPRESENT DIVISION

Component Overview: *Students will receive guided instruction on how to use integer chips to represent division. Students will also learn how to write equivalent division expressions in order to use integer chips. Lastly, students will use what they have learned to create rules for the sign of a quotient.*

TEACHER MOVES

Turn and Talk

Utilize an instructional routine to leverage student-centered instruction for this activity. Students have used integer chips for multiplication, so a Turn and Talk provides them an opportunity to make meaningful connections from multiplication to division.

- How is this the same as multiplication?
- How is it different?

DIFFERENTIATE INSTRUCTION

Consider providing a kinesthetic entry point by having students who are struggling model the problems with concrete integer chips.

7.12.2 GUIDED INSTRUCTION: USING INTEGER CHIPS TO REPRESENT DIVISION (CONTINUED)

ENRICHMENT

- How do the signs of the dividend and divisor relate to the sign of the quotient?
- For what other operation is this true?

3. **Use the pattern of the division in the tables to complete the rules by drawing the sign of the quotient as an integer chip.**

Visual Model of Division	Sign of Quotient
$\frac{+}{+}$	$+$
$\frac{+}{-}$	$-$
$\frac{-}{+}$	$-$
$\frac{-}{-}$	$+$

7.12.3 Collaboration: How Are Integer Multiplication and Division Related?

Jeannie and Solimar were discussing the relationship between $5 \times -2 = -10$ and $(-10) \div (-2) = 5$.

Solimar said, "The multiplication equation can be interpreted as 5 groups of -2, and the division equation can be interpreted as how many groups of -2 are in -10. Isn't that basically the same thing?"

Jeannie replied, "I agree. I also think it is interesting to see how the rules seem to be related. In the multiplication equation, we are multiplying a positive by a negative, and the answer is a negative. In the division equation, we do the reverse of the multiplication problem and find that a negative divided by a negative is a positive."

Solimar said, "Yeah, that's a cool connection. Do you think the rules are the same for multiplication and division?"

Jeannie said, "I think they might be related to each other."

1. Discuss Solimar and Jeannie's conversation with your partner.
   Answers will vary.

2. Create two integer division equations that can be rewritten with the related multiplication equation.
   Answers will vary. Examples shown in table below.

Division Equation	Related Multiplication Equation
$\frac{-14}{7} = -2$	$7 \cdot (-2) = -14$
$\frac{45}{3} = 15$	$3 \times 15 = 45$

7.12.3 COLLABORATION: HOW ARE INTEGER MULTIPLICATION AND DIVISION RELATED?

Component Overview: *Have the students read and discuss the problems with a partner. Students create integer division equations and relate them to integer multiplication equations.*

TEACHER MOVES

Notice and Wonder

Have students discuss what they notice and wonder about the relationship between the equations.

ENRICHMENT

Circulate while students are collaborating and listen for moments to ask questions to extend student thinking.

- What do you notice about the two equations? What do you wonder?
- Why does Jeannie think the rules of multiplication and division are related? Give an example to illustrate this point.
- How can you show that multiplication and division are connected?
- What do you notice about the rules of both multiplication and division?

7.12.4 BRING IT TOGETHER: HOW ARE INTEGER MULTIPLICATION AND DIVISION RELATED?

Component Overview: *Students consolidate their understanding of how multiplication and division are related*

ENRICHMENT

Consider utilizing inquiry to position students for synthesis.

- What do you notice about the factors in each related equation? What do you wonder?
- What do you notice when one factor is negative? What about when both factors are negative?
- How does this connect to what we already know about multiplying integers?

7.12.4 Bring It Together: How Are Integer Multiplication and Division Related?

1. **Write the missing factor in each given equation. Then, write the related division or multiplication equation.**

Given Equation	Can be Thought of as...	Related Equation
$3 \times 7 = 21$	⟺	$21 \div 3 = 7$
$\frac{-48}{4} = -12$	⟺	$(-12)(4) = -48$
$-8 \times -7 = 56$	⟺	$56 \div (-8) = -7$
$20 \div -4 = -5$	⟺	$(-5)(-4) = 20$
$-\left(\frac{36}{4}\right) = -9$	⟺	$-9 \times 4 = -36$

2. **How are rules for multiplication of integers related to the rules for division of integers?**
Answers will vary. The rules for multiplication of integers and the rules for division of integers are the same.

7.12.5 Collaboration: Which One Doesn't Belong?

1. Circle the expression that does not belong with the other three.

$\frac{10}{-5}$ $\frac{-10}{5}$ $\frac{-10}{-5}$ $-\frac{10}{5}$

2. Explain why the expression does not belong.
Answers will vary. Expression 3 does not belong because it is dividing a negative number by a negative number. The quotient of integers with opposite signs is negative. The quotient of integers with the same sign is positive. The value of this expression is positive while the values of the other three expressions are negative.

3. Verify your answers with your partner and, if necessary, resolve any differences.

Quotients can have negative signs in different places.

4. The values $\frac{12}{-2}$, $\frac{-12}{2}$, and $-\left(\frac{12}{2}\right)$ are equivalent. Explain why they are equivalent using your understanding of dividing integers.
Answers will vary. Since the divisor should be positive to indicate the number of groups, $\frac{12}{-2}$ can be rewritten as $\frac{-12}{2}$. Therefore, the two expressions are equivalent. The expression $-\left(\frac{12}{2}\right)$ reads, "the opposite of positive 12 divided into 2 equal-sized groups." The quotient would be equal to 6, and the opposite of 6 is -6. All three expressions are equivalent, as they are all equal to -6.

5. Discuss your explanation with your partner. If necessary, edit your explanation based on your discussion.

6. Create two equivalent expressions for $21 \div (-7)$.

$-\left(\frac{21}{7}\right)$ *or* $\left(\frac{-21}{7}\right)$

7. $\frac{-40}{-8}$ is not equivalent to $-\left(\frac{40}{8}\right)$. Explain why not.
Answers will vary. The resulting sign of the quotient $\frac{-40}{-8}$ is positive because when you divide a negative integer by a negative integer, it always results in a positive integer. The resulting sign of $-\left(\frac{40}{8}\right)$ is negative because it indicates the quotient is the opposite of $\left(\frac{40}{8}\right)$, which is equal to -5.

7.12.5 COLLABORATION: WHICH ONE DOESN'T BELONG?

Component Overview: *Students first work independently to determine and explain which division of integers expression does not belong. Then students discuss their work with a partner to verify their work or make corrections.*

TEACHER MOVES

Which One Doesn't Belong?

Monitor for students with different responses and elicit their responses whole group to orient students to each other and the concept of equivalence.

ENRICHMENT

- Why does the expression you selected not belong?
- How do you know what the sign of the quotient will be?
- Are any of the expressions equivalent?

DIFFERENTIATE INSTRUCTION

- Why does the expression you selected not belong?
- How do you know what the sign of the quotient will be?
- Are any of the expressions equivalent?

ENRICHMENT

- What do you know about the quotient when the dividend is positive, and the divisor is negative? Are these expressions equivalent?
- How do you know?
- How does this compare to multiplying integers with the same sign?

7.12.6 GUIDED PRACTICE: DIVIDING INTEGERS

Component Overview: *Students practice dividing integers. Consider providing integer chips, if needed.*

QUESTIONING

For students who finish early, consider having them create the related multiplication equations for each division equation. To add complexity, consider replacing some of the integers in each division equation with more complex integers.

ENRICHMENT

How do you know the expressions are equivalent using quotient rules?

7.12.6 Guided Practice: Dividing Integers

1. Find each quotient.

   a. $12 \div -4 = -3$

   b. $\frac{-63}{-9} = 7$

   c. $-\left(\frac{28}{4}\right) = -7$

   d. $60 \div 5 = 12$

   e. $-\left(\frac{-30}{-5}\right) = -6$

2. Create two expressions equivalent to $\frac{-20}{10}$.

   $\frac{20}{-10}$ *and* $-\left(\frac{20}{10}\right)$

3. What is the value of the expressions above?

   -2

7.12.7 Your Turn!

1. In each column below, create three different division expressions that would result in the given quotient.
Answers will vary.

-4	10	-7
1. $\frac{-40}{10}$	1. $\left(\frac{100}{10}\right)$	1. $\left(\frac{70}{-10}\right)$
2. $-\left(\frac{12}{3}\right)$	2. $\left(\frac{-20}{-2}\right)$	2. $\left(\frac{-84}{12}\right)$
3. $\frac{16}{-4}$	3. $-\left(\frac{-50}{5}\right)$	3. $-\left(\frac{-35}{-5}\right)$

7.12.7 YOUR TURN!

Component Overview: *Students work independently to create three different division expressions to represent a given quotient. Consider modeling an example before having students work through the table on their own. After students are finished working, have them work in pairs to verify their responses before sharing responses whole group.*

QUESTIONING

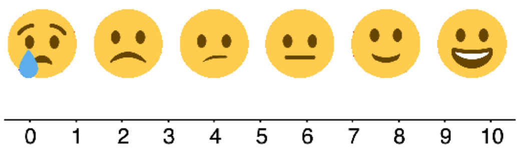
For students who may finish early, consider challenging them to create more division expressions that would result in the given quotients for each column.

DIFFERENTIATE INSTRUCTION

Display an anchor chart for students who are struggling. When determining the equivalent expressions, prompt students to determine the sign of the quotients first and then divide the integers.

7.12.8 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with understanding and using the rules for multiplying and dividing integers.
Answers will vary.



7.12.8 WRAP-UP: REFLECTION

Component Overview: *Students complete a reflection indicating their confidence in using rules to multiply and divide integers.*

DIFFERENTIATE INSTRUCTION

If time allows, ask students to justify their emoji rating. They could write an explanation summarizing the rules. They can also write and solve a few sample problems to demonstrate their knowledge and skills.